MONASH University

When Algorithms Choose Who Gets Help

Week 5 — Search, consolidated

Julian Garcia

FIT1061 — Introduction to Artificial Intelligence

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MILESTONE

Haiti earthquake

The birth of crisis mapping

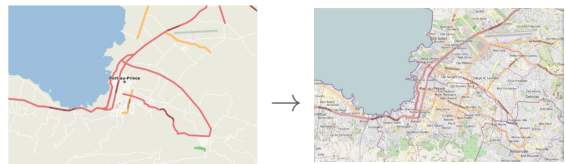
magnitude 7.0 · Port-au-Prince · 12 January 2010

MILESTONE

Haiti earthquake

The birth of crisis mapping

magnitude 7.0 · Port-au-Prince · 12 January 2010



before — near-blank

after — dense



Maps: OSM Haiti progression (Maron / OSM). Photo: M. Dormino / UNDP, Port-au-Prince, 12 Jan 2010.

What happened?

The gap. Before the quake, OpenStreetMap held only part of Port-au-Prince's roads — other web maps, even less. Responders needed accurate maps to route logistics.

The build. Satellite firms released before/after imagery. Volunteers traced roads, buildings, damage, and camps — so the map could **guide the response**.

What happened?

The gap. Before the quake, OpenStreetMap held only part of Port-au-Prince's roads — other web maps, even less. Responders needed accurate maps to route logistics.

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- Satellite imagery → **OpenStreetMap**
- ∼**53 h**: 400+ edits
- ∼**1 month**: ∼600 mappers
- More detailed than the **UN's & Google's** road data

Haiti launched a mapping movement

Two jobs: **build the map**, then **route over it**.

1

BUILD THE MAP

Volunteers trace the roads

Humanitarian OpenStreetMap Team (HOT) · Missing Maps

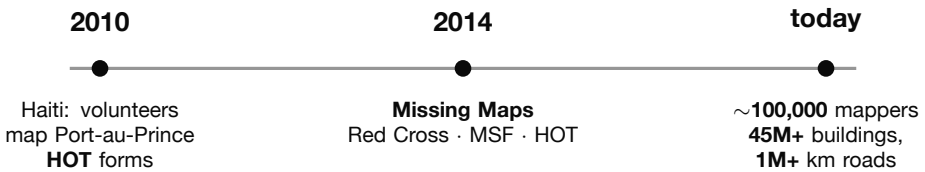
2

ROUTE OVER IT

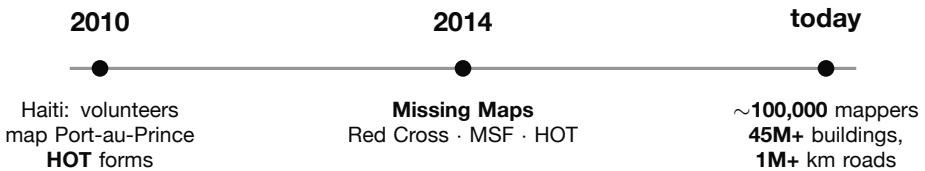
Software plans the trucks

e.g. the World Food Programme's *Route The Meals*

The movement it became



The movement it became



Agencies then **route over** these maps.

The UN's **World Food Programme** plans deliveries with *Route The Meals* — **US$3.1M** saved across 10 countries.

The search is neutral machinery.

The search is neutral machinery.

The values live in three choices:

what we **map**

(the graph)

what we **follow**

(the heuristic)

what we **measure**

(the objective)

What we map is a choice
What we follow is a choice
The Pareto frontier
What we measure is a choice
In the lab

What we map is a choice

So far — one kind of question



One **start**, one **goal**, find a path.

So far — one kind of question



One **start**, one **goal**, find a path.

We've built **three** ways to do it — three tools, one neutral job.

Recall — three searches

Breadth-first

Explore in **layers**; the **waiting list** is a **queue** (first in, first out). Guarantees the **fewest-edges** path.

Trades **memory** for a **guarantee**.

Recall — three searches

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Depth-first

Dive **deep** along one branch; the **waiting list** is a **stack** (last in, first out). Finds *a* path fast — **no** shortest guarantee.

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Recall — three searches

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Trades the **guarantee** for **memory**.

Greedy best-first

Follow the **heuristic**: expand whatever *looks* closest to the goal. Fast — but can be **misled**.

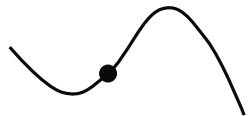
Trades **guarantees** for **speed**.

Two kinds of problem



Path-finding

There's a start and a goal. You want the **route**.
(breadth-first, depth-first, greedy)



Optimisation / local search

No goal state. You hold a **candidate** and improve it.
(hill climbing, n-queens)

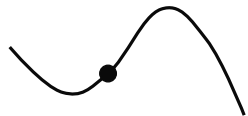
First question on any problem: **which kind is this?**

Two kinds of problem



Path-finding

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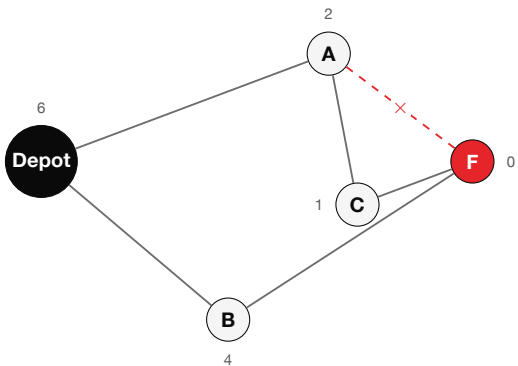
Optimisation / local search

No goal state. You hold a **candidate** and improve it.
(hill climbing, n-queens)

First question on any problem: **which kind is this?**

Today's is **path-finding** — one start, one goal.

The humanitarian network



Depot → F

A **destroyed** road is a missing edge (dashed).

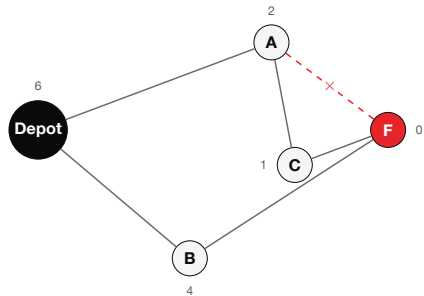
grey number = straight-line estimate to F.

Someone drew this map.

Breadth-first on this map

Breadth-First Search

```
function BFS(start, goal):
    frontier ← Queue()
    frontier.enqueue(start)
    visited ← {start};   came_from ← {}
    while frontier is not empty:
        current ← frontier.dequeue()
        if current = goal: return path
        for each neighbour of current:
            if neighbour not in visited:
                visited.add(neighbour)
                came_from[neighbour] ← current
                frontier.enqueue(neighbour)
```

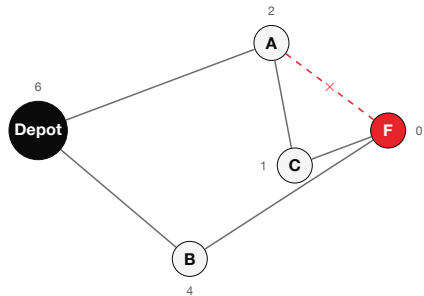


FIFO queue — explores in layers, ignores the estimate.

Greedy best-first on this map

Greedy Best-First Search

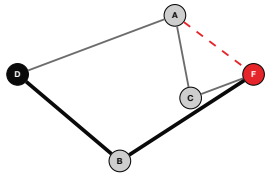
```
function Greedy(start, goal):
    frontier ← PriorityQueue by h(n)
    frontier.add(start)
    visited ← {start};   came_from ← {}
    while frontier is not empty:
        current ← frontier.pop_lowest_h()
        if current = goal: return path
        for each neighbour of current:
            if neighbour not in visited:
                visited.add(neighbour)
                came_from[neighbour] ← current
                frontier.add(neighbour)
```



Priority by estimate h — always expand the lowest-h node.

Only the **frontier** changes from BFS.

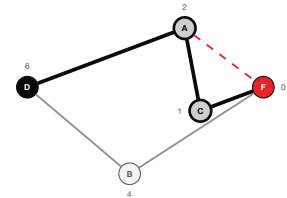
One map, two searches, two routes



Breadth-first

Fans out in layers. Ignores the hint.

Finds the 2-edge route: D → B → F



Greedy best-first

A *looks* closest ($h = 2$) — but the map has **no** A → F road. The map, not the estimate, forces the detour.

Detours: D → A → C → F (3 edges)

Same estimates — **the map decides the route.**

What we map is a choice



Port-au-Prince, before the quake

We don’t choose the **terrain**.

We choose what goes on the **map**.

The search routes only over what’s on the map — so a place left off it **gets no help**.

What we map is a choice

What we follow is a choice

The Pareto frontier

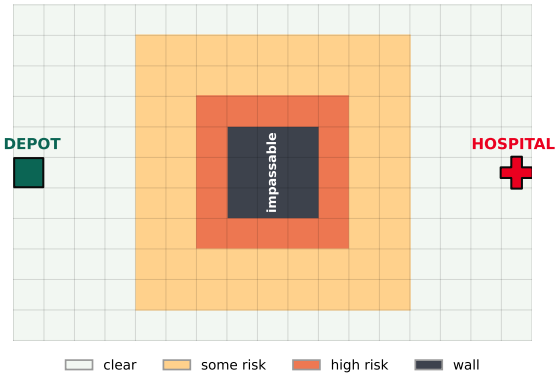
What we measure is a choice

In the lab

What we follow is a choice

Through the hazard, or around?

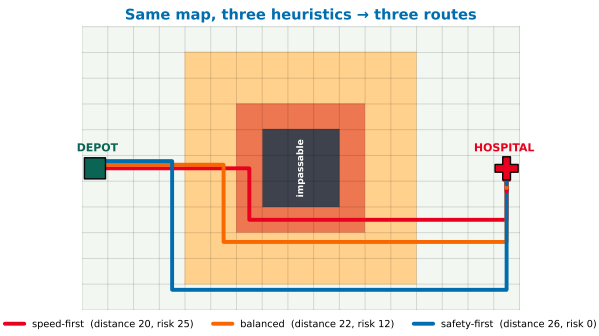
The terrain: depot → hospital, hazard in orange



One convoy, **depot** → **hospital**.

Greedy takes whatever *looks* best — and **“best”** is **your** definition.

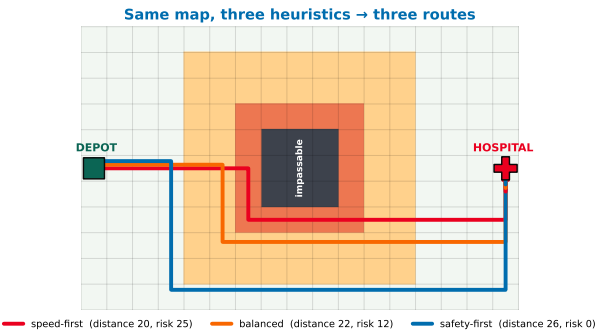
Same map, three priorities



Same map, same greedy. Change what we tell it to **care about**:

- **speed-first** — ignore risk.
- **balanced** — some caution.
- **safety-first** — avoid the hazard.

Same map, three priorities

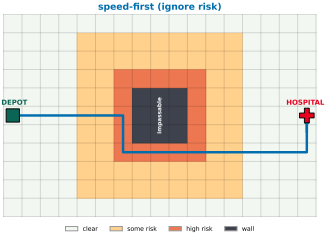


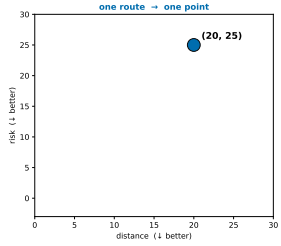
Same map, same greedy. Change what we tell it to **care about**:

- **speed-first** — ignore risk.
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One priority, one route — **three routes**.

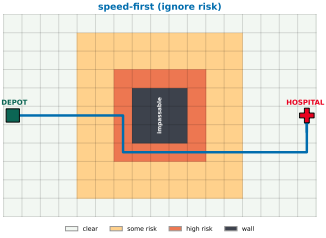
How we score a route

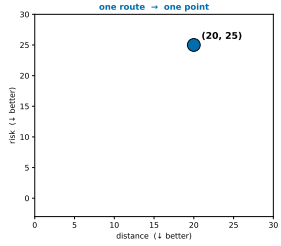




- **distance** — the number of **steps**.
- **risk** — add up the risk of **every cell** the route crosses (clear **0** · low ring **1** · high ring **3**).

How we score a route





- **distance** — the number of **steps**.
- **risk** — add up the risk of **every cell** the route crosses (clear **0** · low ring **1** · high ring **3**).

Two numbers → one **point**: here **(20, 25)**.

Each route, two numbers

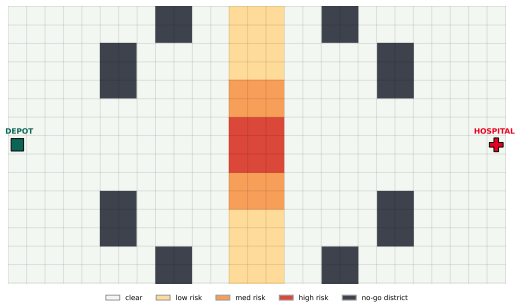
Route	distance (↓)	risk (↓)
speed-first (through the hazard)	20	**25**
balanced (close to the edge)	22	12
safety-first (around)	26	**0**

Each route, two numbers

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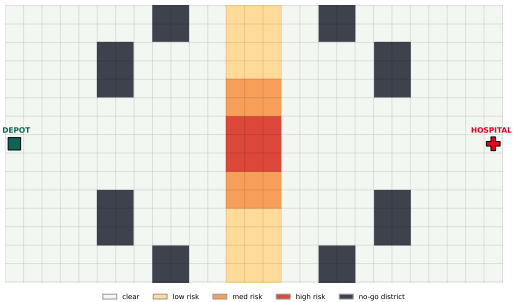
Each route is a **point**.

Now scale up — a real city



No-go **districts** (collapsed, flooded) and **risk zones** of different intensity.

Now scale up — a real city

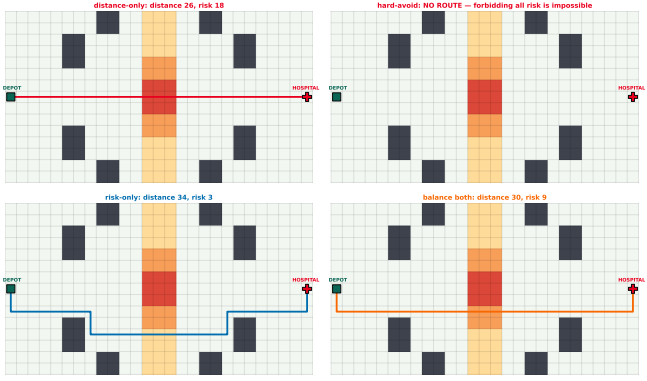


No-go **districts** (collapsed, flooded) and **risk zones** of different intensity.

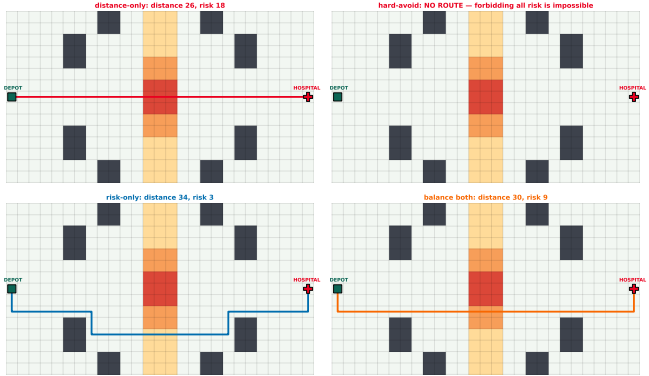
You **can't** eyeball the best route now.

Let's try the obvious **rules** and watch each one.

Four rules, four outcomes



Four rules, four outcomes



Each rule chases **one thing** and pays on the other — so we must **weigh both**. **How do we say that** to a computer?

You can’t forbid risk — only trade it

Key Takeaway

“Never enter any risk” is a **hard rule**: it rejects a cell no matter what.

But if **every** route to the hospital crosses some risk, a hard rule leaves **no route at all**.

You can’t forbid risk — only trade it

Key Takeaway

“Never enter any risk” is a **hard rule**: it rejects a cell no matter what.

But if **every** route to the hospital crosses some risk, a hard rule leaves **no route at all**.

So we don’t **forbid** risk. We **weigh** it — and choose how much.

Designing a heuristic

So give every step a **combined score**: how far the goal is, *plus* a penalty for the risk here.

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$$h(n) = \underbrace{\text{distance to goal}}_{\text{get there}} + \lambda \cdot \underbrace{\text{risk here}}_{\text{stay safe}}$$

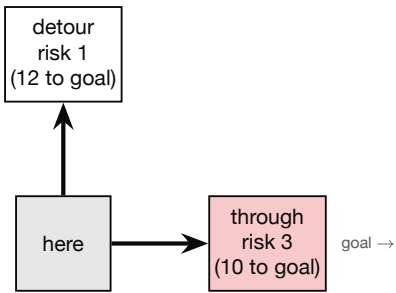
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$$h(n) = \underbrace{\text{distance to goal}}_{\text{get there}} + \lambda \cdot \underbrace{\text{risk here}}_{\text{stay safe}}$$

λ — the **safety weighting** — is how much distance you'll spend to avoid one unit of risk.

Greedy at the junction



At the edge of the hazard, greedy compares two steps with $h = \text{distance} + \lambda \cdot \text{risk}$:

- **through** — closer (10), but risk 3.
- **detour** — farther (12), but risk 1.

Where does greedy **switch** from through to detour?

Poll

How many extra steps would *you* spend to avoid one unit of risk? That number is your λ.

Pick your safety weighting λ:

- **0 — speed above all**
- **about 1 — some caution**
- **about 2–3 — lean safe**
- **4 or more — safety first**

Vote now — scan the code for your campus, *or* type the address.



Melbourne

pollev.com/juliangarcia



Malaysia

pollev.com/onghueyfang767

Your λ builds your route

The three routes we started with were just **three values of** λ:

Your λ builds your route

The three routes we started with were just **three values of** λ:

$$\text{speed-first} = \lambda\,0 \quad \cdot \quad \text{balance} = \lambda\,1 \quad \cdot \quad \text{safety-first} = \lambda\,4$$

Slide the dial, and the route **moves with it**.

What we map is a choice

What we follow is a choice

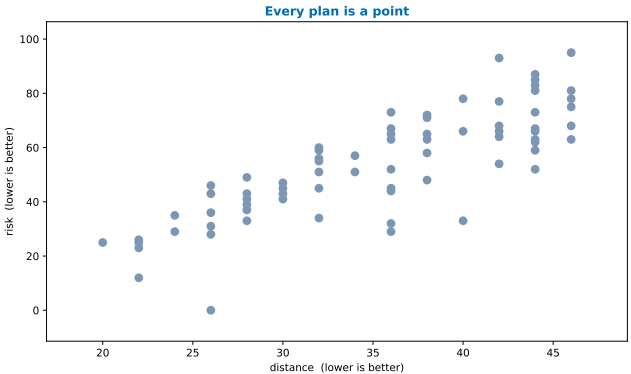
The Pareto frontier

What we measure is a choice

In the lab

The Pareto frontier

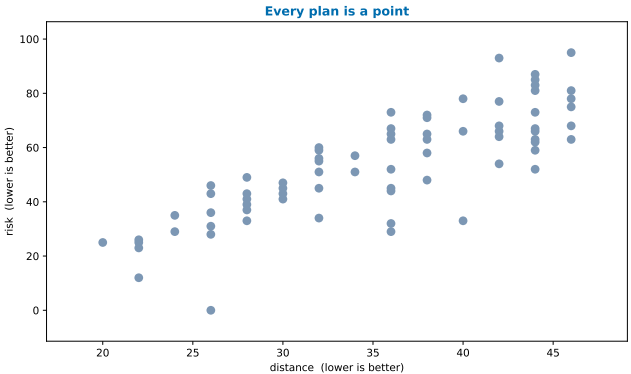
Every plan is a point



Score every possible route — a whole **cloud** of plans.

Some are just **worse** than another — beaten on *both* counts.

Every plan is a point

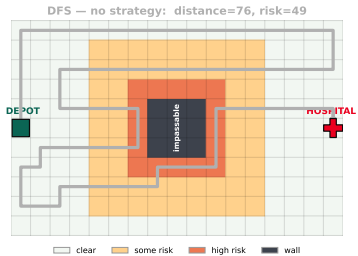


Score every possible route — a whole **cloud** of plans.

Some are just **worse** than another — beaten on *both* counts.

Which ones can we **rule out**?

Pareto dominance



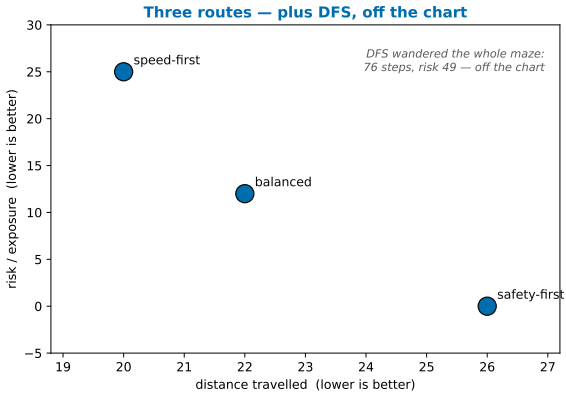
Depth-first, no strategy: a route beaten on *both* counts.

Dominance and the Pareto frontier

Option A **dominates** option B if A is *at least as good* on every axis and *strictly better* on at least one. An option is **Pareto-optimal** if nothing dominates it. The set of Pareto-optimal options is the **Pareto frontier**.

Named for **Vilfredo Pareto**, an economist who, around 1900, studied how resources spread across a society — the *frontier* is the outer edge of what's achievable.

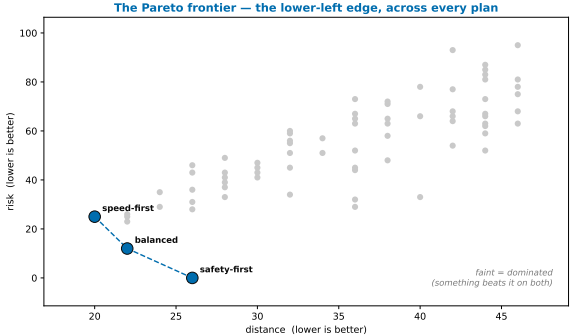
Eliminate the dominated ones



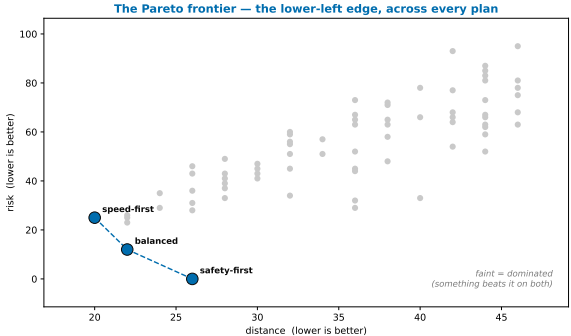
Can't cross off a whole cloud by hand — so take your **three routes**, plus **DFS** at **(76, 49)**, off the chart.

- Take each route. Is there another that is **no worse on either axis** and **strictly better on at least one** (fewer steps or less risk)?
- If yes — it's dominated. **Rule it out.**
- Which survive — and which is beaten on *both* counts?

The Pareto frontier



The Pareto frontier



The **lower-left edge** survives — the frontier.

The same edge your **three routes** traced — now across *every* plan.

What the frontier does — and does not

Key Takeaway

The frontier tells you the trade-offs you **can’t beat**.

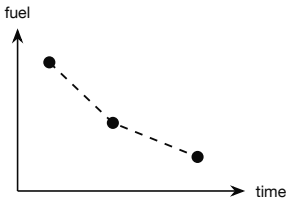
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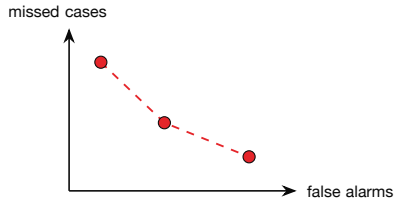
It does **not** tell you which one to pick.

You've seen this frontier before



Your navigation app

fastest · fuel-saving · fewer highways



Coming later

when the axes are **people**

Same shape every time — and someone always picks the point.

What we map is a choice

What we follow is a choice

The Pareto frontier

What we measure is a choice

In the lab

What we measure is a choice

who decides?

Picking a route puts one axis first

- “distance” is a stand-in for **speed to the patients waiting**.
- “risk” is a stand-in for **danger to the convoy crossing the hazard**.

```
how much distance to avoid one unit of risk?
```

Picking a route puts one axis first

- “distance” is a stand-in for **speed to the patients waiting**.
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```
how much distance to avoid one unit of risk?
```

That safety weighting is the **value choice** — made *before* the search runs.

Safety, or reach?

REACH first

send the convoy **through** — people are waiting

the crew takes the risk

SAFETY first

route **around** — don’t drive into danger

the patients wait

Safety, or reach?

REACH first

send the convoy **through** — people are waiting

the crew takes the risk

SAFETY first

route **around** — don't drive into danger

the patients wait

Both sit **on the frontier** — the search ranks neither above the other.

Which one to send is a **value judgement** the search can't settle.

Who bears the risk?

SETS the weighting

whoever runs the operation

often far from the road

≠

BEARS the risk

the crew in the convoy

… and the patients who wait

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Should whoever **sets** the weighting be the one who **bears** it?

Poll

The map that decided who got help first was drawn by online volunteers.

Whose needs are most likely to be missing from that map?

- **People with fast internet**
- **People far from any mapped road or device**
- **Large aid organisations**
- **No one is missing**

Vote now — scan the code for your campus, *or* type the address.

Melbourne
pollev.com/juliangarcia

Malaysia
pollev.com/onghueyfang767

Whose voice is missing?

Discussion — back to Haiti

- Who chose **what went on the map** — and who was far from a keyboard?
- Who set **what the search follows** (how heavily it weighs risk)?
- Who chose **what we measure** as success — speed, or safety?
- If the route goes wrong, **who answers** — can it be “the algorithm”?

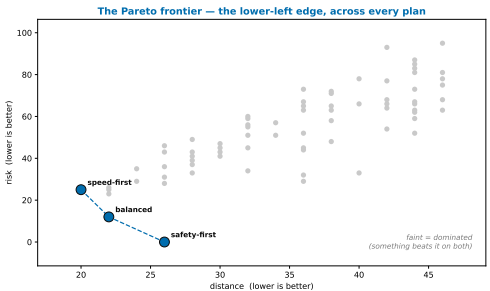
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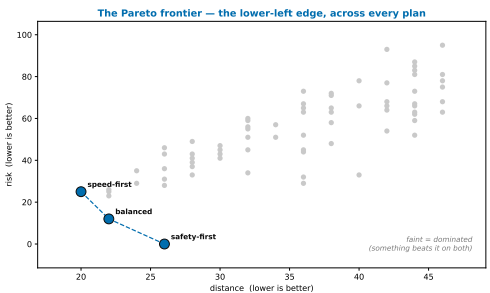
A map built to guide aid still reflects the **priorities** of whoever drew it.

A harder axis



The risk score counts every person's **exposure the same**. But the same exposure harms a **vulnerable** person more.

A harder axis



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Should the risk axis **weight the vulnerable more**? Would the frontier **move**?

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The search only carries them out.

The choosing, and the responsibility for it, are **ours**.

What we map is a choice

What we follow is a choice

The Pareto frontier

What we measure is a choice

In the lab

In the lab

In the lab this week

Pass tier. **Aiming for a Pass**

1. Run breadth-first, depth-first, and greedy on the humanitarian network.
2. Run greedy under **different heuristics** and plot the routes it builds.
3. Find the Pareto frontier over those routes.
4. Reason about which axis you chose — and why.

In the lab this week

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Credit tier. **Aiming for a Credit** — all of the above, plus:

- Compare the three searches across **different graph structures**.
- Make a stuck search reliable with **random restarts**.

In the lab this week

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Credit tier. **Aiming for a Credit** — all of the above, plus:

- Compare the three searches across **different graph structures**.
- Make a stuck search reliable with **random restarts**.

Distinction tier. **Aiming for a Distinction** — all of the above, plus:

- What could a cheap, **never-pessimistic** estimate of remaining cost look like — and how would it change the search?

Key takeaways

- What we **map** is a choice — search routes over the graph we build, not the terrain. On the Haiti map, the gaps were whoever was far from a keyboard.

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Key takeaways

- What we **map** is a choice — search routes over the graph we build, not the terrain. On the Haiti map, the gaps were whoever was far from a keyboard.
- What we **follow** is a choice — how heavily we weight risk builds a different route.
- What we **measure** is a choice — the frontier finds the routes we can't beat, but not which to pick. That's a human's call.